

Prevalence and Priorities: Characterizing Challenging Behaviors across Caregiver and Clinician Report Measures to Inform Intervention Targets in Children with Syndromic IDD

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Background

- Behavior rating scales are part of best practices for evaluating outcomes in RCTs. However, these measures may not capture behaviors specific to an individual or caregiver treatment priorities.¹
- Web Intervention for youth with Genetic Syndromes (WINGS)** is a behavioral intervention RCT designed for syndromic IDD populations that utilizes both behavior rating scales and patient-reported outcome measures to inform intervention target behaviors.
- We used a **multimethod approach**, including the behavior checklist, **Behavior Problems Inventory (BPI-01)**, with the **Parent Top Problems (PTP)**, a family-centered, open-ended interview, to develop representative outcomes for trials in autism and IDD populations.^{2,3}

Objectives

- Characterize **challenging behaviors** that inform intervention goals in the cohort using both the **PTP**, a caregiver-centered measure, and the **BPI-01**, a behavior rating scale.
- Identify **similarities and differences between the PTP** (patient-centered outcome) and **BPI-01** (severity/frequency scale) in characterizing challenging behavior profiles.

Participants

37 caregivers of eligible participants with syndromic IDD completed the BPI-01 during clinical screening and the PTP with an Independent Evaluator (IE) during baseline according to established protocols (See Table 1 for Measure Overview).

Child Demographics

- Age:** 2-12 years, $M = 6.68$, $SD = 3.25$
- Sex assigned at birth:** Male 75.68%, $n = 28$; Female 24.32%, $n = 9$
- Co-occurring autism:** 48.68%, $n = 18$
- Genetic Syndrome**
 - Angelman Syndrome: 16.22%, $n = 6$
 - Fragile X Syndrome: 21.62%, $n = 8$
 - Phelan-McDermid Syndrome: 29.73%, $n = 11$
 - Other: 32.43%, $n = 12$
- ID Level**
 - Borderline ID: 5.41%, $n = 2$
 - Mild ID: 45.95%, $n = 17$
 - Moderate ID: 37.84%, $n = 14$
 - Severe ID: 10.81%, $n = 4$

Table 1. PTP and BPI-01 Measure Overview

	PTP	BPI-01
Description	Semi-structured clinician elicitation of parent target problems. ²	52-item respondent-based behavior rating measure across three subscales: 1) self-injurious, 2) stereotyped, and 3) aggressive/destructive behavior. ³
Outcome	Each target problem is given a label and narrative describing the topography, frequency, context, and impact of the behavior. ²	Each behavior is rated on frequency (<i>never, monthly, weekly, daily, hourly</i>) and severity (<i>mild, moderate, severe</i>), yielding a frequency and severity score for all three subscales. ³
Behavior Examples	Aggression: Aggression, face grabbing/scratching, biting Compliance Issues: Noncompliance, difficulty with/resisting transitions Safety Concerns: Elopement, awareness/safety	Stereotyped Behavior: Repetitive body movements, manipulating objects Self-Injurious Behavior: Hitting head, teeth grinding, self-biting Aggressive/Destructive Behavior: biting, scratching, grabbing/pulling³

Figure 1. PTPs: Caregiver Reports of High-Priority Challenging Behaviors for Intervention Targets

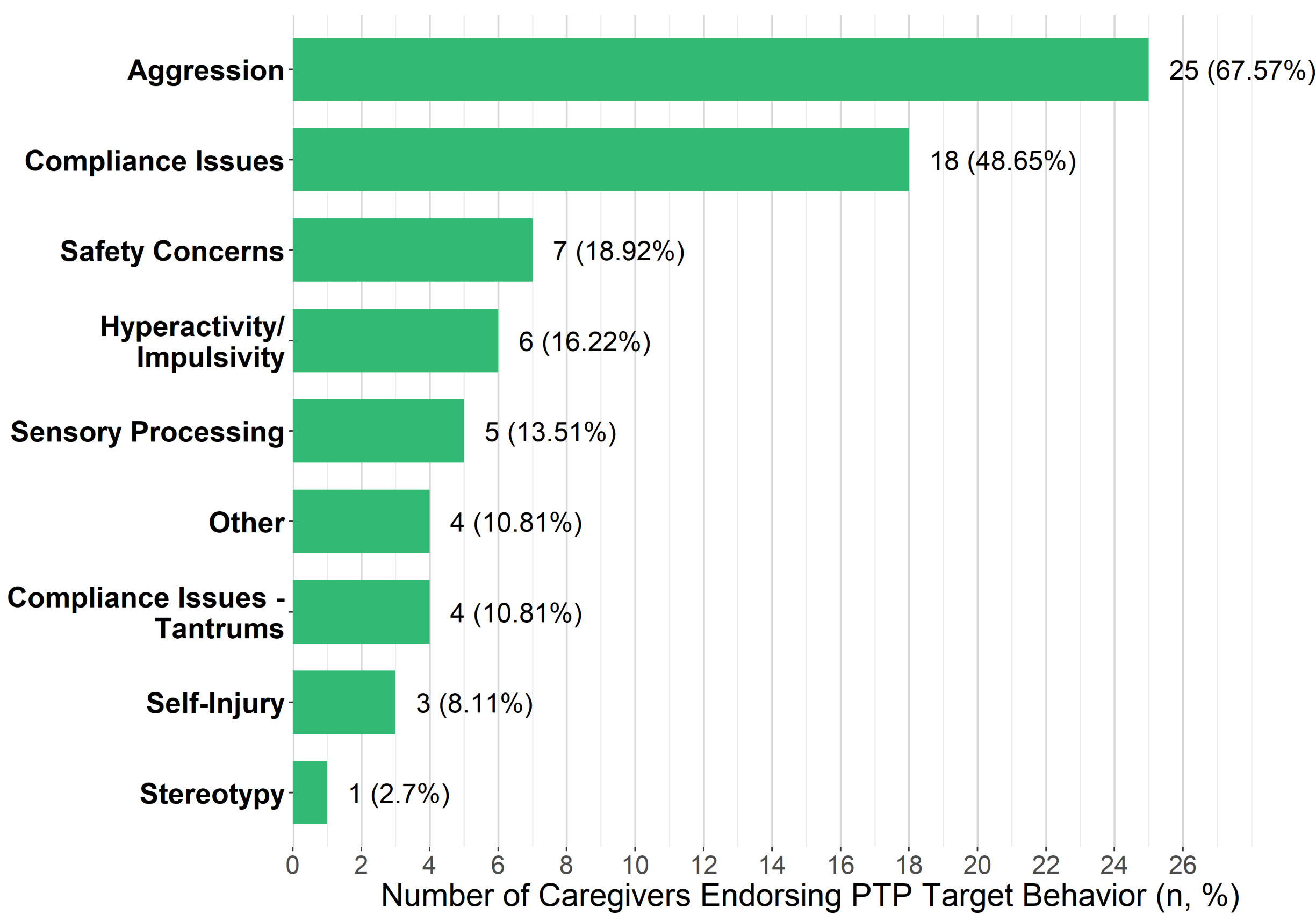
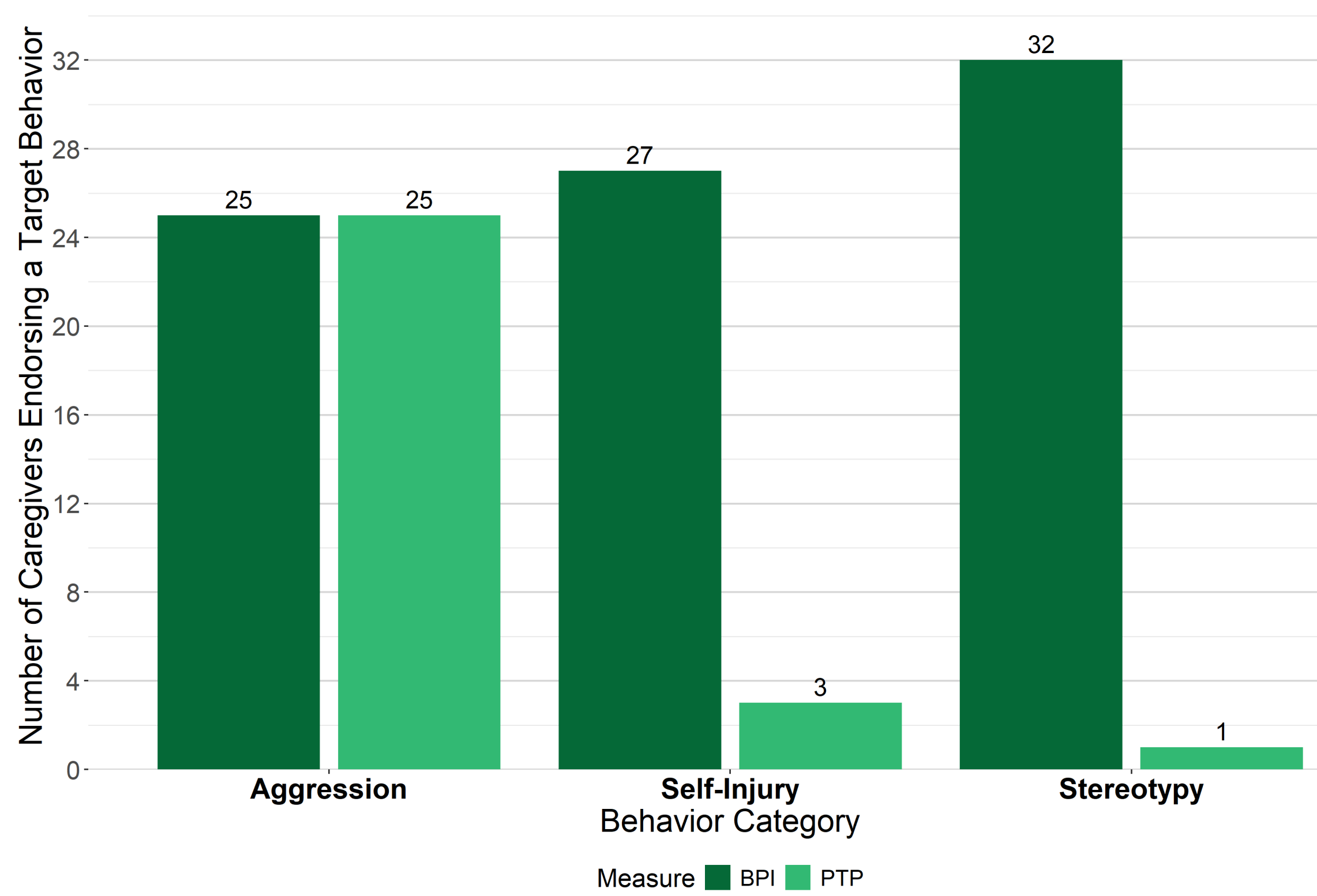


Table 2. BPI-01: Caregivers Endorsing Challenging Behaviors by Frequency and Severity Criteria

BPI Subscale	≥1 behavior on a Daily/Hourly Basis	≥1 behavior on a Daily/Hourly Basis <u>also</u> rated as Moderate or Severe
	Caregivers n (%)	
Aggressive/Destructive Behavior	25 (67.57%)	25 (67.57%)
Self-Injurious Behavior	27 (72.97%)	24 (64.86%)
Stereotyped Behavior	32 (86.49%)	29 (78.38%)

Figure 2. Shared Topographies Across the PTP and BPI-01



Procedure

- 76 PTP labels** were categorized into 9 topography categories using methodology in previous RCTs¹ (Figure 1).
- Initial agreement on PTP categorization was 75% and reviewers reviewed the remainder to reach 100% agreement.
- We examined endorsement of Aggression, Stereotypies, and Self-Injury using the following criteria:
 - Endorsement of **at least 1 target problem on the PTP**
 - Endorsement of **at least 1 behavior which occurs on Daily or Hourly basis on the BPI-01**⁴ (Table 2, Column 3)

Results

- Patient-reported outcomes:** The **top three** most frequently endorsed challenging behaviors on the **PTP** were **Aggression** (67.57%, $n = 25$), **Compliance Issues** (48.65%, $n = 18$), and **Safety Concerns** (18.92%, $n = 7$) (Figure 1).
- Rating scale:** When examining caregivers who endorsed **≥1 behavior on a Daily/Hourly basis on the BPI-01**, 67.57% caregivers ($n = 25$) endorsed Aggressive/Destructive, 72.97% ($n = 27$) endorsed Self-Injurious, and 86.49% ($n = 32$) endorsed Stereotyped behaviors (Table 2, Column 3).
- Rating scale:** When examining caregivers who endorsed **≥1 behavior on a Daily/Hourly basis also rated Moderate or Severe on the BPI-01**, endorsement on the Aggressive/Destructive subscale was the same, but slightly lower for the Self-Injurious and Stereotyped Behavior subscales (Table 2, Column 3).
- Shared topographies:** 67.57% ($n = 25$) caregivers endorsed **Aggression** on the PTP and BPI-01, while patterns across **Self-Injury** and **Stereotypy** differed (Figure 2).

Discussion

- While certain topographies, such as **Stereotypies** and **Self-Injury**, may be **highly prevalent** amongst autism and IDD populations, our data suggest that these **are not treatment priorities for caregivers**.
- Differences in the top behavior topography endorsed on the PTP (Aggression) and BPI-01 (Stereotypy) in our cohort demonstrate **the value of multimethod approaches** that include open-ended measurement along with standardized measurement tools to develop **meaningful, patient-centered treatment outcomes**.



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ClinicalTrials.gov NCT06139172



References